Chaotic Dynamical Systems and the Logistic Map

A Complete Mathematical Reference

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1. What Is Chaos?

The word “chaos” in everyday language implies complete randomness and disorder. In mathematics, it means something more precise and, in some ways, more surprising: a deterministic system that produces behavior so sensitive to initial conditions that it is effectively unpredictable over long time horizons, despite following exact rules at every step.

The distinction from true randomness is crucial. A chaotic system produces sequences that look random when you examine them statistically — they pass tests for uniform distribution, low autocorrelation, and long periods — but they are generated by a completely deterministic algorithm. Given exactly the same initial conditions, a chaotic system will reproduce exactly the same sequence. The apparent randomness arises not from any true stochasticity but from the exponential amplification of tiny initial differences, a property called **sensitive dependence on initial conditions**.

This makes chaotic systems useful as **pseudo-random number generators**: they produce deterministic but statistically random-looking sequences at very low computational cost, with no external randomness required.

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2. The Logistic Map

The simplest and most studied chaotic system is the **logistic map**, a one-dimensional recurrence relation:

$$x_{t+1} = r, x_t (1 - x_t)$$

For $x_t \in [0, 1]$ and $r \in [0, 4]$, this map keeps all iterates within $[0, 1]$. The parameter r controls the system’s behavior, and the transition from orderly to chaotic behavior as r increases is one of the most beautiful phenomena in dynamical systems theory.

The term $x_t(1 - x_t)$ is a **logistic function** — it is zero at both extremes ($x = 0$ and $x = 1$), achieves its maximum of $1/4$ at $x = 1/2$, and is concave throughout. Multiplied by r, it produces a parabola with maximum $r/4$. For the map to keep all values within $[0, 1]$, we need the maximum to not exceed 1, which requires $r \leq 4$.

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3. The Bifurcation Diagram

The behavior of the logistic map as a function of r reveals a cascade of complexity, summarized in the **bifurcation diagram**. Plotting the long-run values of x_t as a function of r shows four distinct regimes.

$r \in [0, 1]$: Extinction. All initial conditions converge to $x = 0$. The fixed point $x^* = 0$ is stable.

$r \in (1, 3)$: Stable fixed point. All initial conditions converge to the positive fixed point $x^* = 1 - 1/r$. The system is simple and predictable.

$r \in (3, 3.57...)$: Period doubling. At $r = 3$, the fixed point loses stability and the system begins oscillating between two values (period-2 cycle). At $r \approx 3.449$, the period-2 cycle loses stability and is replaced by a period-4 cycle. This **period-doubling cascade** continues at smaller and smaller intervals of r, with periods $2, 4, 8, 16, \ldots$ accumulating at the **Feigenbaum point** $r_\infty \approx 3.5699$.

$r \in (3.57, 4]$: Chaos. Above the Feigenbaum point, the system is generically chaotic — except for narrow windows where periodic orbits briefly appear before giving way to chaos again. At $r = 4$, the chaos is full and the map is equivalent to a coin-flipping process in a precise mathematical sense.

3.1 The Feigenbaum Constants

The period-doubling cascade is not specific to the logistic map. Any smooth one-dimensional map with a single quadratic maximum undergoes the same bifurcation sequence, with the ratio of successive period-doubling intervals converging to the **first Feigenbaum constant**:

$$\delta = \lim_{n\to\infty} \frac{r_n - r_{n-1}}{r_{n+1} - r_n} \approx 4.6692...$$

This constant is universal across an entire class of maps — it does not depend on the specific form of the map, only on the fact that the maximum is quadratic. This universality, discovered by Mitchell Feigenbaum in 1975, was a stunning discovery: a numerical constant appearing identically in systems as different as fluid turbulence, chemical oscillations, and electronic circuits.

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4. Sensitivity to Initial Conditions and Lyapunov Exponents

The hallmark of chaos is exponential sensitivity to initial conditions. If two initial conditions differ by a tiny amount ϵ, their trajectories diverge as:

$$|x_t - x_t’| \approx \epsilon, e^{\lambda t}$$

where λ is the **Lyapunov exponent**. When $\lambda > 0$, nearby trajectories diverge exponentially — the system is chaotic. When $\lambda < 0$, they converge — the system is stable. When $\lambda = 0$, the system is on the boundary between stability and chaos (typically at period-doubling bifurcation points).

For the logistic map at $r = 4$, the Lyapunov exponent is exactly $\lambda = \ln 2 \approx 0.693$. This means the separation between two initially nearby trajectories doubles every time step — after n steps, any initial difference has been amplified by 2^n. After 50 steps, a difference of 10^{-15} (at the limit of double precision floating point) has been amplified to order 1, making the sequences completely uncorrelated. This is why the logistic map at $r \approx 4$ is an effective pseudo-random number generator.

The Lyapunov exponent of the logistic map can be computed as the time-average of the log-derivative:

$$\lambda = \lim_{T\to\infty} \frac{1}{T}\sum_{t=0}^{T-1} \ln|f’(x_t)| = \lim_{T\to\infty} \frac{1}{T}\sum_{t=0}^{T-1} \ln|r(1 - 2x_t)|$$

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5. Statistical Properties of the Chaotic Regime

At $r = 4$, the logistic map has a remarkable analytical property. The substitution $x_t = \sin^2(\pi \theta_t)$ transforms the map into:

$$\theta_{t+1} = 2\theta_t \pmod 1$$

which is simply the **doubling map** — a standard source of pseudo-randomness. This means the invariant distribution of x_t under the $r = 4$ logistic map is the **arcsine distribution** (also called the Chebyshev measure):

$$p(x) = \frac{1}{\pi\sqrt{x(1-x)}}$$

This distribution is supported on $(0,1)$, symmetric around $1/2$, and places more weight near the endpoints 0 and 1 than near the center — the opposite of a uniform distribution. For timing jitter applications, this is useful: the logistic map at $r \approx 4$ produces delays that are preferentially near the extremes of the allowed range (very short or very long), with intermediate delays being less common.

The autocorrelation of the chaotic sequence at $r = 4$ is zero at all non-zero lags — the values at successive time steps are uncorrelated, even though the sequence is completely deterministic. This is what makes it useful as a timing jitter source: it passes statistical tests for randomness while requiring only a few arithmetic operations to generate the next value.

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6. The Chaotic Timing Engine

The application of the logistic map to timing jitter is as a **chaotic timing engine**. Rather than using a uniform pseudo-random number generator to sample inter-event delays, you use the logistic map iterate x_t to generate delays:

$$\text{delay}_t = \frac{\text{base_ms} \cdot (0.5 + x_t)}{1000}$$

The $(0.5 + x_t)$ term shifts the range from $[0, 1]$ to $[0.5, 1.5]$, centering the delays around the base delay. The division by 1000 converts from milliseconds to seconds. The result is a deterministic but statistically unpredictable sequence of delays with:

$$\mathbb{E}[\text{delay}] = \frac{\text{base_ms} \cdot (0.5 + \mathbb{E}[x])}{1000} = \frac{\text{base_ms} \cdot (0.5 + 0.5)}{1000} = \frac{\text{base_ms}}{1000}$$

since $\mathbb{E}[x] = 0.5$ under the arcsine invariant measure.

The variance of the delay is:

$$\text{Var}[\text{delay}] = \frac{\text{base_ms}^2}{10^6} \cdot \text{Var}[x] = \frac{\text{base_ms}^2}{10^6} \cdot \frac{1}{8}$$

since $\text{Var}[x] = 1/8$ under the arcsine distribution ($\mathbb{E}[x^2] = 3/8$, $(\mathbb{E}[x])^2 = 1/4$).

The chaotic engine requires only one floating-point multiplication and one subtraction per step — far cheaper than a general pseudo-random number generator — and produces delays with the statistical properties needed to defeat periodic pattern detection.

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7. Connection to Information Theory

There is a deep connection between chaos, randomness, and information theory captured by the concept of **metric entropy** (also called Kolmogorov-Sinai entropy or measure-theoretic entropy). For a dynamical system, the entropy measures how quickly information about the initial condition is “lost” as the system evolves — equivalently, how quickly new information is generated.

For the logistic map at $r = 4$, the metric entropy equals the Lyapunov exponent: $h = \lambda = \ln 2$ bits per iteration. This means the map generates one bit of new (unpredictable) information per iteration — exactly the information-theoretic equivalent of a fair coin flip. It cannot generate more than this (that would require a “super-chaotic” system) but it generates exactly this amount, which is optimal for a system evolving on the unit interval.

This connection to entropy is not coincidental. The Pesin identity ($h = \sum_{\lambda_i > 0} \lambda_i$ for multidimensional systems) establishes a general relationship between metric entropy and positive Lyapunov exponents. Chaos generates information at a rate determined by how quickly nearby trajectories diverge — a beautiful unification of dynamical systems theory and information theory.

See also: Concurrency Design Patterns; Acquisition Functions in Bayesian Optimization